Pokemon VGC Matchup Predictor

Predicting Competitive Doubles Battle Outcomes Using Machine Learning

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Introduction

- Competitive Pokemon VCG doubles battles involve two players selecting Pokemon to compete against each other. Before the battle begins, both players select two pokemon to lead their team.
- Our objective for this project is to develop a supervised machine learning model that predicts the winner of a Pokemon VGC doubles battle using only the two leading Pokemon from each player, alongside stats of the pokemon and other additional information about the Pokemon.
- Team matchups can influence the outcome of a battle. Starting Pokemon provide information about the matchip before any moves are even made.

Research Questions

RQ1. To what extent can the two starting Pokemon of each player predict the winner of a Gen 9 VGC doubles match?

VGC Battle Logs Data Source

- Hugging Face: cameronangliss/vgc-battle-logs
- Pokemon Showdown Battle logs.
- Gen 9 VGC Champions format
- Regulations M-A and M-B

VGC Battle Logs Data Collection & Processing

- Collected Pokemon Showdown battle logs
- Extracted relevant information from each battle
- Cleaned and processed the raw data
- Isolated the starting pokemon and battle outcomes for model development

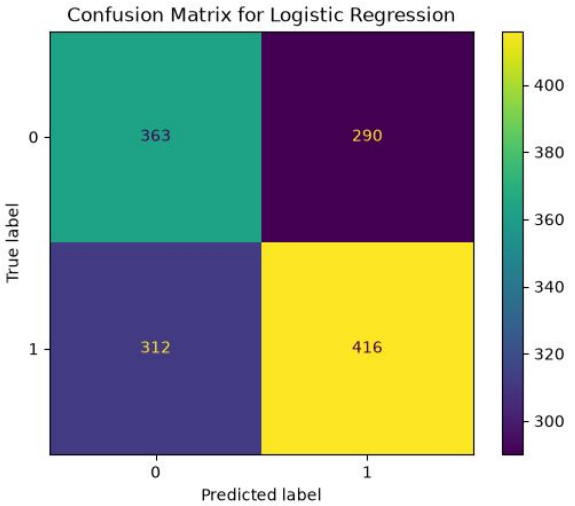
Regulation M - A ~ 50 MB in raw data & ~6.5MB in processed data

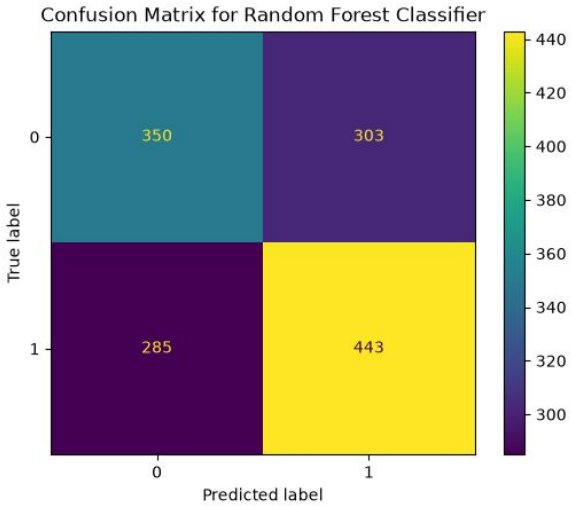
Regulation M - B ~ 2.3 MB in raw data & 300KB in processed data

Methodology

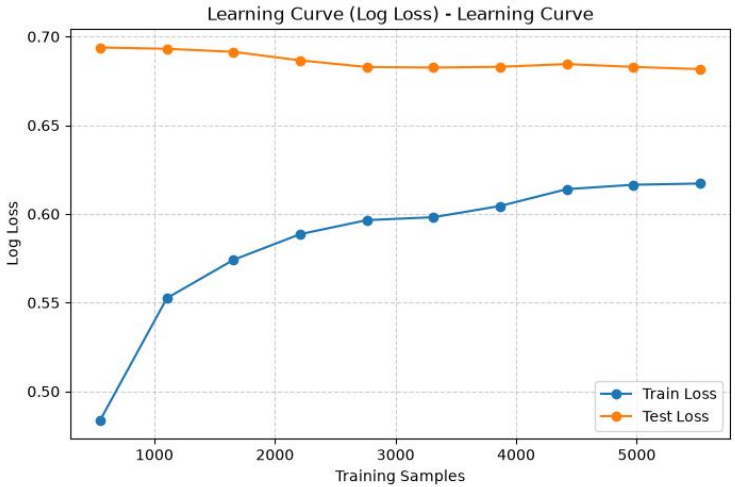
- Inputs:
  - Player 1’s two leading Pokemon
  - Player 2’s two leading Pokemon
- Prediction:
  - A percentage of how likely either side can win
- Models:
  - Majority Classifier
  - Logistic Regression

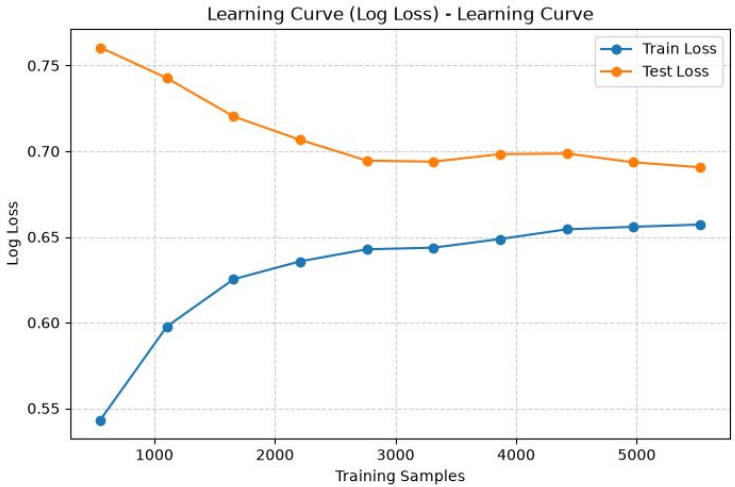
Confusion Matrix



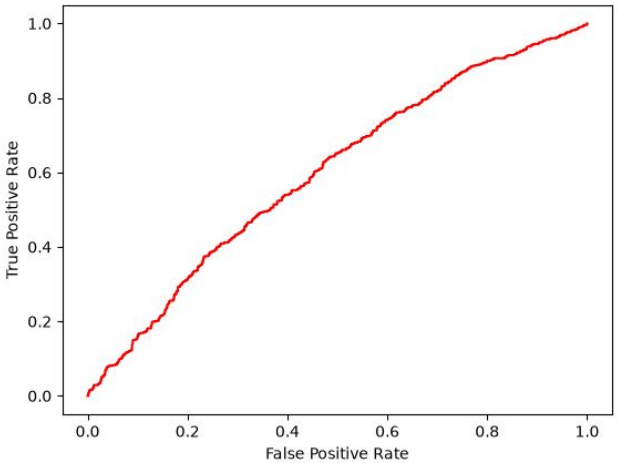


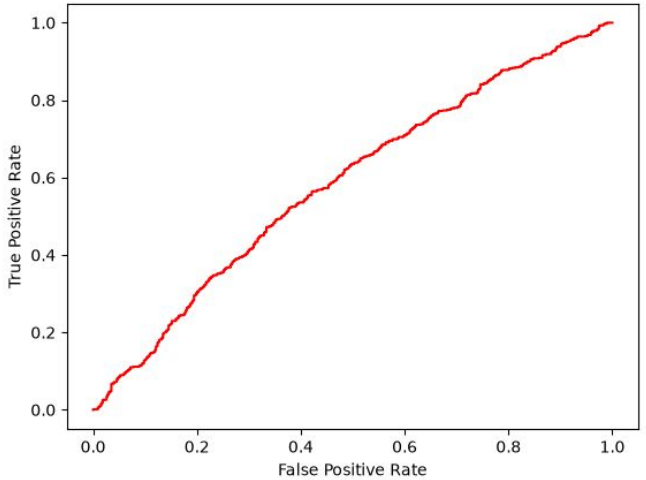
Learning Curve





ROC Curve


True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0


True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

Conclusion

- Developed a supervised machine learning model for predicting Pokemon VGC doubles battle outcomes.
- Implemented Logistic Regression and Majority Classification baseline to evaluate predictive performance
- Starting Pokemon alone may not capture the full potential of the battle.